



16 — Shallow State: `$state.raw`

PERFORMANCE REACTIVITY

Q. How would you handle a massive array of 10,000 data points in Svelte 5 without causing severe memory and performance overhead?

ANSWER. Normally, `$state` builds a heavy 'proxy' around every single nested property of an object to track microscopic changes. But if you have 10,000 data points for a financial chart, that tracking overhead will freeze the browser. You use `$state.raw` to completely skip deep tracking, guaranteeing blazing fast renders as long as you swap the whole array out at once.

```
<script>
  // Perfect for thousands of data points
  let data = $state.raw([
    { x: 1, y: 10 },
    { x: 2, y: 15 }
  ]);

  function loadNewData(newData) {
    // You MUST replace the whole array.
    // data.push() will not trigger updates.
    data = newData;
  }
</script>
```

ChartData.svelte

`$state`

DEEP PROXY

Object 1
(Tracked)

Object 2
(Tracked)

Object 3
(Tracked)

VS

`$state.raw`

SHALLOW

Object 1
(Ignored)

Object 2
(Ignored)

Object 3
(Ignored)

① **Summary.** Deep reactivity is expensive on massive datasets. This matters because `$state.raw` lets you opt out of that

overhead. You get blazing fast renders for large lists as long as you swap the whole array instead of mutating individual items.